

Geist in the Machine — Volume Spine

A Research Atlas of Dialectical Agent Pedagogy

Liam Magee*

Education Policy, Organization and Leadership
University of Illinois Urbana-Champaign

April 2026

Geist in the Machine

A Research Atlas of Dialectical Agent Pedagogy

Spine generated from `paper-full-2.0.md` v3.0.119. The statuses and the evidence map below are projected from `atlas.yaml` and stay in sync by construction.

Orientation

This atlas is a *projection* of the canonical paper (`paper-full-2.0.md`) into a short spine plus a set of semi-autonomous modules. The paper remains the single source of truth — every paper claim the modules derive from is machine-checked against the evaluation DB under provable discourse. The atlas exists so the work can be read by mechanism family rather than as one ~280-page argument, and so live arcs (drama machine, charisma, ontology) can grow as modules without restructuring the whole.

The consolidated single PDF is still available and unchanged: it *is* the source. Build it with `npm run atlas:consolidated`. The modules below are derived views of that same source — they cannot drift from it, because they are slices of it.

Theoretical map

The programme asks: *what kinds of agent-internal structure produce better pedagogical relation, and which apparent structures are illusory, redundant, or too costly to pay?* Three mechanism families organise the answer — **recognition** (calibration + conditional error

*This sentence is the only one actually authored by Liam Magee in this paper.

correction, the settled results), **drama** / **charisma** (voice, persona, productive internal conflict), and **ontology** / **state** (whether a modelable learner interior exists at all).

Methodological map

The apparatus is itself a contribution: process tracing adapted for agent architectures, a provable-discourse framework that machine-verifies every mechanism claim against the evaluation DB, four rubric iterations treated as construct development, and the test suite as analytical provenance.

Evidence map

Each module declares a claim-status (grammar at the foot of this spine). The table is generated from the manifest, so a status change here propagates to the spine automatically.

#	Module	Family	Status
1	Recognition as Calibration	Recognition and Pedagogical Calibration	SETTLED
2	Superego as Conditional Error Corrector	Recognition and Pedagogical Calibration	SETTLED
3	The Adaptive-Responsiveness Null	Recognition and Pedagogical Calibration	KILLED
4	Provable Discourse and Process Tracing for Agent Architectures	Apparatus as Method	METHODS
5	Drama Machine Pedagogy	Drama and Charisma	PLANNED
6	The Id-Director and Charismatic Pedagogy	Drama and Charisma	EXPLORATORY
7	Ontology, State, and the Failure of Rich Learner Interiors	Ontology, State, and the Limits of Interior Modeling	KILLED
8	Rubrics as Instruments and Optimization Signals	Apparatus as Method	METHODS
9	Human Learning Translation Agenda	Translation and Future Programme	FUTURE

The modules

Module 1 — Recognition as Calibration

[SETTLED] Settled within current evidence · *Recognition and Pedagogical Calibration*

The clean prompt-level mechanism: recognition acts as calibration, a floor-lifting and variance-narrowing transformation of tutor output. Includes the A10/A10b matched-pedagogy controls that locate the effect at the orientation-family level.

Sources: §3.2, §6.1, §7.9

Module 2 — Superego as Conditional Error Corrector

[SETTLED] Settled within current evidence · *Recognition and Pedagogical Calibration*

The architecture-level mechanism: what the superego catches, how critique resolves across deliberation rounds, the factorial interaction, and the ~15–17% additivity deficit consistent across three models.

Sources: §6.2, §6.4, §7.3

Module 3 — The Adaptive-Responsiveness Null

[KILLED] Killed / closed under a specified gate · *Recognition and Pedagogical Calibration*

The third pre-registered mechanism, turned from a failed hypothesis into a substantive well-powered null — including the insight-action gap and the A16 cumulative-rewrite pre-registration.

Sources: §6.3

Module 4 — Provable Discourse and Process Tracing for Agent Architectures

[METHODS] Methods / apparatus contribution · *Apparatus as Method*

The apparatus-as-method module: superego-critique taxonomy, the provable-discourse framework that verifies every mechanism claim against data, and the reproducibility chain.

Sources: §5.3, §5.9, §5.11, §7.4

Module 5 — Drama Machine Pedagogy

[PLANNED] Planned — scaffold, prose pending · *Drama and Charisma*

An active arc. Develops deliberative tension, role differentiation, and anti-sycophantic friction as a pedagogical architecture extending — but not reducible to — the ego/superego result. Prose lands as a new section of the canonical paper; this module is the scaffold.

Sources: — (scaffold; prose pending)

Module 6 — The Id-Director and Charismatic Pedagogy

[EXPLORATORY] Exploratory · *Drama and Charisma*

A parallel architecture family — a transient ego repeatedly authored by a durable id — scored on a Weberian charisma rubric independent of the v2.2 tutor rubric.

Sources: §6.7, §8.9

Module 7 — Ontology, State, and the Failure of Rich Learner Interiors

[KILLED] Killed / closed under a specified gate · *Ontology, State, and the Limits of Interior Modeling*

The deepest theoretical arc, mostly negative: the adaptive runner and trap methodology, bilateral-ToM and state-schema ablations, hypothesis tracking, and the concealed-interior signal test. Asks whether there is a modelable interior at all, or whether interiority is an artifact of trace design.

Sources: §6.8, §6.9, §6.10

Module 8 — Rubrics as Instruments and Optimization Signals

[METHODS] Methods / apparatus contribution · *Apparatus as Method*

Treats the four rubric iterations, the measurement paradox, scoring, and automated prompt tuning as instrument design rather than incidental tooling.

Sources: §5.2, §5.6, §7.7, §7.8

Module 9 — Human Learning Translation Agenda

[FUTURE] Future empirical agenda · *Translation and Future Programme*

Separates judged tutor quality from human learning outcomes, motivated by the model-dependent tutor-learner asymmetry and the synthetic-learner / LLM-judge limits. The translation programme and its pilot infrastructure.

Sources: §6.5, §8.1, §8.2

Claim-status grammar

- [SETTLED] — Settled within current evidence
- [SCOPE-BOUND] — Supported but scope-bound
- [EXPLORATORY] — Exploratory
- [KILLED] — Killed / closed under a specified gate
- [SPECULATIVE] — Speculative / theoretical
- [METHODS] — Methods / apparatus contribution
- [PLANNED] — Planned — scaffold, prose pending
- [FUTURE] — Future empirical agenda